



The DNA of work

AI Job Market Impact Tracker

User manual

How the dashboard works, what every term means,
and where every number comes from.
Grounded in the Anthropic Observed Exposure methodology.
Six regions - weekly refresh.

Developed by AWA

Advanced Workplace Associates | Research instrument

1. What this tool is

The AI Job Market Impact Tracker is a research instrument that turns disparate signals about AI's effect on labour markets into a single, weekly-refreshed evidence base, viewable as a dashboard and downloadable as raw data. It covers six regions: the United States, United Kingdom, India, European Union, Asia Pacific, and Australia.

The dashboard opens in a **Simple** view built for a fast, plain-language read: one AI Pressure Index per region, four pillar signals, and a sector pulse ranking ten industry sectors by AI pressure. A **Deep** toggle exposes everything underneath — the ten raw KPIs with measured/modelled provenance, the analytical charts, and "under the hood" lineage panels that show every input, calibration band and weight behind every derived number. Sections 5.1 and 5.2 explain both layers in detail.

It exists because the public debate about AI and jobs lurches between two extremes — nothing is changing, or everything is. The reality, as Anthropic's own research shows, is that the early signals are subtle, narrow, and easy to miss in aggregate statistics. The tracker is built to detect those signals as they emerge.

1.1 Who it is for

Workplace strategy consultants, workforce planners at large employers, policy researchers, and journalists who need an accountable, sourced view of how the labour market is shifting.

1.2 What it is not

- It is not an oracle. It cannot predict which individual will lose a job.
- It is not a causal claim. It tracks correlations and attributes events to AI only where the employer or a credible source says so.
- It is not investment or hiring advice. It is a research tool.

2. Background research — what the methodology is built on

Massenkoff and McCrory (Anthropic, 5 March 2026) published a paper titled Labor market impacts of AI: A new measure and early evidence. It introduced a metric they called **Observed Exposure**, designed to measure not just which jobs are theoretically vulnerable to AI but which are already, in practice, being done by AI in real-world traffic.

2.1 The three inputs

Observed Exposure combines three independent data sources:

- **O*NET**: the US Department of Labor's database of around 800 occupations, each decomposed into specific tasks.
- **Eloundou et al. (2023)**: an academic paper that scored every O*NET task on a 0 / 0.5 / 1 scale (called β) for its theoretical exposure to LLM speed-up.
- **The Anthropic Economic Index**: real-world usage data showing what tasks Claude is actually being asked to do, and whether the use is automated (AI does the work) or augmentative (human and AI collaborate).

2.2 The headline findings (US, as of March 2026)

- 97% of observed Claude task usage falls into categories Eloundou et al. rated as theoretically feasible ($\beta \geq 0.5$).
- The most exposed US occupations are **Computer Programmers (75%)**, **Customer Service Representatives**, **Data Entry Keyers (67%)**, and **Financial Analysts**.
- 30% of US workers have zero exposure under this measure – their tasks appeared too infrequently in Claude usage to count.
- BLS 2024–2034 employment projections fall by 0.6 percentage points for every 10 pp of additional Observed Exposure.
- Top-quartile-exposure US workers are more likely to be female (+16 pp), more educated, older, and earn 47% more than the unexposed group.
- Aggregate unemployment in the top-quartile-exposure group has not increased significantly since late 2022, though the job-finding rate for workers aged 22 to 25 in exposed occupations has dropped about 14%.

Massenkoff and McCrory's central message is one of analytical modesty: AI's labour-market impact may be like the internet or trade with China rather than like the COVID shock. If so, it will not be obvious from headline unemployment numbers. A measure that tracks the narrowing gap between theoretical capability and observed deployment is more likely to detect the inflection early.

3. How Observed Exposure is calculated

The score is built bottom-up from tasks to occupations to occupation categories.

3.1 Task level

Every O*NET task receives three flags:

- β from Eloundou et al.: 0, 0.5, or 1.
- **Observed usage**: does this task appear with sufficient frequency in the Anthropic Economic Index, and is it work-related?
- **Use pattern**: of the observed usage, what share is automation versus augmentation?

A task's coverage equals 1 if $\beta \geq 0.5$ and it has sufficient work-related Claude usage, weighted by 1.0 for automation and 0.5 for augmentation. Otherwise, coverage = 0.

3.2 Occupation level

An occupation's Observed Exposure is the time-weighted average of its task-level coverage scores. Time weights come from O*NET's reported time-on-task data: tasks workers spend more of their day on count more.

3.3 Sector and regional aggregates

Occupation scores are aggregated to occupation categories (e.g., Computer & Math, Office & Admin) by current employment weight. For regions other than the US, the local occupation classification is crosswalked to O*NET-SOC first; where local time-on-task data exists, it replaces the US weights.

Robustness check. The Anthropic paper varies several judgement calls (the β coding, the automation weight, the usage threshold) and reports that the Spearman rank correlation of occupation-level exposures across reasonable parameterisations is very high. The ranking is stable even when the absolute numbers move.

4. The KPI catalogue

The dashboard shows ten KPIs per region. Each KPI has a definition, a formula or source, a refresh cadence, and an alert band that triggers escalation in the weekly digest.

AI-attributed layoffs (YTD)

Definition	Cumulative layoffs since 1 January where the employer cited AI. Measured for the US from the Challenger, Gray & Christmas monthly Job Cut Report; other regions remain modelled pending a comparable source.
Source	Challenger, Gray & Christmas monthly report (US, measured); Layoffs.fyi tracker as context.
Refresh	Monthly report, checked weekly.
Alert band	More than 500 in a single week.

Early-career unemployment delta (proxy)

Definition	Youth minus overall unemployment rate, in percentage points, seasonally adjusted. A documented proxy for the top-quartile-exposure cohort: statistics agencies do not publish unemployment by AI exposure, so the original Massenkoff & McCrory diff-in-diff can only be replicated from CPS microdata. Measured: US 20-24 (BLS), UK 18-24 (ONS), EU under-25 (Eurostat), AU 15-24 (ABS).
Source	BLS CPS, ONS LMS, Eurostat une_rt_m, ABS Labour Force; Massenkoff & McCrory 2026 framing.
Refresh	Monthly.
Alert band	More than 0.5 pp rise in the differential over 13 weeks.

22-25 hire rate

Definition	Year-on-year change in the monthly job-start rate among workers aged 22 to 25 in exposed occupations.
Source	BLS CPS panel, ONS LFS, equivalent for each region. Brynjolfsson, Chandar & Chen (2025) provide the methodology template.
Refresh	Monthly.
Alert band	More than 10% YoY drop.

AI-mention posting share

Definition	Percentage of job postings whose description references AI, ML, GenAI, prompt engineering, or related skills.
Source	Indeed Hiring Lab feeds, supplemented by Naukri JobSpeak (India) and Seek (Australia).
Refresh	Weekly.
Alert band	Doubling YoY.

Capability gap

Definition	Theoretical β (Eloundou) minus Observed Exposure, averaged across the region's occupations weighted by current employment. Expressed in percentage points.
Source	Eloundou et al. 2023 plus Anthropic Economic Index.
Refresh	Monthly.
Alert band	Gap closes by more than 3 pp.

Augmentation share

Definition	Percentage of Claude conversations in the region's traffic mix classified as collaborative (augmentative) rather than end-to-end (automated).
Source	Anthropic Economic Index quarterly release.
Refresh	Quarterly.
Alert band	Below 45% or above 60%.

Exposed-occupation posting index

Definition	Mean Indeed postings index across eight high-exposure sectors, seasonally adjusted, indexed to February 2020 = 100. Measured for the US, UK, EU (DE+FR mean) and AU; India and APAC use Adzuna exposed-category counts (indexed to their first measured week) once the API key is configured.
Source	Indeed Hiring Lab job postings tracker (open GitHub dataset); Adzuna categories for India/APAC.
Refresh	Weekly.
Alert band	More than 5% drop on a 4-week rolling basis.

AI-skill salary premium

Definition	Median advertised salary for postings mentioning AI minus median for comparable non-AI postings, expressed as a percentage of the non-AI median.
Source	Indeed Hiring Lab plus Lightcast (formerly Burning Glass).
Refresh	Monthly.
Alert band	Premium widens or narrows by more than 10%.

Graduate posting

Definition	Year-on-year change in entry-level postings in exposed roles. Big 4 graduate disclosures (KPMG, Deloitte, EY, PwC) are incorporated where published.
Source	Indeed Hiring Lab, IFOW graduate study, employer filings.
Refresh	Quarterly.
Alert band	More than 15% YoY drop.

Net AI-attributed job creation

Definition	New AI, ML, data, security postings minus AI-attributed displacement events over the latest 12 months, in thousands of roles.
Source	WEF Future of Jobs 2025 regional baseline plus posting-derived adjustments.
Refresh	Quarterly.
Alert band	Net swings into negative territory.

5. How to read each dashboard panel

5.1 Simple and Deep views – the derived layer (added Aug 2026)

The dashboard opens in **Simple** view: one **AI Pressure Index** (0–100) per region and four plain-language pillar signals – Job displacement (are jobs being cut?), Hiring pullback (are exposed roles being hired less?), Early-career squeeze (are young workers feeling it first?), and AI adoption pace (how fast is AI entering work?). Each pillar scores its input KPIs against fixed calibration bands (linear 0–100 between a documented floor and ceiling, clamped), then combines them with fixed weights; the composite weights the pillars 30/25/25/20%. Status words – Low (<25), Moderate (25–50), Elevated (50–70), High (≥70) – always accompany the colour, and every score carries a confidence chip: the share of its weight backed by measured (not modelled) sources that week.

Switching to **Deep** view reveals everything underneath: the ten raw KPI tiles with their measured/modelled badges, the full analytical charts, and an "Under the hood" panel on every pillar card tracing each input's raw value → calibration band → normalised score → weight → contribution, with source links. The derived layer is computed in the browser from `current.json` and `historical.csv` – the lineage displayed is the live calculation, so the dashboard and its documentation cannot drift apart. Bands, weights and the missing-input rule are specified in `DERIVED_METRICS.md` in the repository. Scores compare a region to itself over time and are not cross-country exposure rankings, consistent with the quartile-rank rule in section 8.3.

5.2 Sector pulse (added Aug 2026; all six regions)

AI impact by sector – banking & financial markets, insurance, IT & software, telecom & media, manufacturing, healthcare, retail, professional services, education, government. Each sector's **AI exposure index** is the employment-share-weighted mean of the locked Anthropic Observed Exposure scores across that sector's occupation mix (occupation×industry matrices: BLS OEWS for the US; ONS SOC×SIC ad-hoc joined to the AWA UK model; Eurostat ISCO×NACE; ILOSTAT ISCO×ISIC annual matrices for Australia, India and the APAC composite, which pools Singapore, Japan and Korea), displayed within-region only with the top sector indexed at 100. **Sector pressure** blends exposure (50%) with the 12-week Indeed sector posting trend (30%) and official vacancy/employment momentum (20%) on fixed bands. Where a region's statistics only publish industry sections, sectors sharing a section (e.g. banking and insurance in section K) share an exposure score and are marked "section-level". Weekly signals and quarterly matrix rebuilds run in CI; every signal carries the measured/modelled badge, and Deep view exposes the full lineage. Taxonomy concordance: `model/sector_concordance.csv` in the repository.

5.3 Panel-by-panel guide (Deep view)

Most exposed occupations table

Ten occupations in the selected region with the highest Observed Exposure score. The blue bar is the score itself, 0 to 100%. The red bar is the capability gap – how much room the score has to grow before reaching theoretical β ceiling. When the red bar shrinks week on week without the blue bar staying flat, that is the earliest signal AI is moving deeper into that occupation.

Net job change (created vs displaced)

A horizontal bar chart of jobs created (green) and displaced (red) over the last 12 months, by occupation family, in thousands of roles. Displacement figures filter to attribution-confidence of 3 or higher, so vague "AI-driven" press claims do not dominate. Use the source & method tab on the card to see which buckets the figures roll up from.

Demographics in the exposed quartile

A doughnut chart of the age distribution of workers in the top-quartile-exposure occupations. The chart subtitle shows the female/male split. Compare across regions to see how exposure concentrates differently – for example, the Indian top-quartile skews far younger than the European one.

Augmentation vs automation

A doughnut chart showing the share of Claude usage in the region that is collaborative versus end-to-end. A high augmentation share suggests workers are still in the loop; a high automation share suggests AI is doing more of the work alone. Anthropic's March 2026 release shows augmentation just over half of Claude.ai conversations.

Posting volume in exposed roles

A 12-week line of the exposed-occupation posting index (February 2020 = 100, seasonally adjusted). A downward slope of more than 5% over a 4-week rolling window is the alert band.

Capability gap by sector

Twin bars per sector: blue is theoretical β , red is Observed Exposure. The space between is headroom. This is the chart to watch over months and quarters – if red moves toward blue in a sector, that sector is converting capability into deployment.

News & events feed

Most recent AI-attributed events in the region. Each is tagged with an attribution-confidence score: 5 = company filing or direct statement; 4 = first-party data such as Indeed or Naukri; 3 = credible reporting; 2 = secondary reporting; 1 = unsourced. Filter mentally to confidence ≥ 3 unless you are specifically tracking the narrative.

6. Regional adaptations

Observed Exposure was calibrated on US O*NET tasks and US Claude traffic. Three adaptations let it work consistently across the six regions:

- **Crosswalking:** the local occupation classification is mapped to O*NET-SOC. UK uses SOC2020 (ONS crosswalk), India uses NCO-2015 (Nasscom crosswalk), the EU uses ISCO-08 (CEDEFOP crosswalk), Australia and New Zealand use ANZSCO 2022.
- **Time-use reweighting:** where local time-on-task data exists, it replaces the US weights. For India, this is rare; for the EU, Eurostat LFS provides partial coverage.
- **Country brief substitution:** Anthropic has published country briefs for India and Australia and is rolling out more. Where a brief exists, its task-mix is used in place of global Claude data.

Important caveat for cross-region reading. Anthropic country briefs measure what Claude users in that country are doing – not what the country's workforce is doing. Cross-country comparisons should always use the within-country quartile rank as the primary comparable; absolute coverage is shown alongside as context only.

Region	Classification	Statistical baseline	Real-time signal
United States	O*NET-SOC native	BLS CPS, JOLTS, EP-2024-34	Indeed Hiring Lab, Lightcast, layoffs.fyi
United Kingdom	SOC2020 → O*NET	ONS LFS, Vacancy Survey, IFOW	Indeed Hiring Lab UK, LinkedIn Economic Graph UK
India	NCO-2015 → O*NET	NSO PLFS, Nasscom Strategic Review	Naukri JobSpeak weekly, Foundit
European Union	ISCO-08 → O*NET	Eurostat LFS, Cedefop Skills Forecast	EURES, LinkedIn EU Economic Graph
Asia Pacific	Mixed (SSOC, JSCO, KSCO)	Singapore MOM, Japan MHLW, Korea KOSIS	JobStreet, Jobsdb, Wantedly
Australia	ANZSCO 2022	ABS Job Vacancies, Treasury	Indeed Hiring Lab AU, Seek Employment Trends

7. Data sources and refresh cadence

Sources are tiered by reliability and latency. Lower tiers are used to colour and accelerate, never to overwrite higher-tier numbers.

Tier	Sources	Latency	Role
Tier 1 — Methodological anchor	Anthropic Economic Index releases; Anthropic country briefs; Hugging Face dataset Anthropic/EconomicIndex.	Quarterly	Sets Observed Exposure per occupation.
Tier 2 — Official statistics	US BLS, UK ONS, Eurostat, India NSO, Australia ABS, Singapore MOM.	1-3 months	Baseline employment, unemployment, demographics.
Tier 3 — Real-time market	Indeed Hiring Lab (postings + AI tracker), Adzuna API, Naukri JobSpeak, Singapore MOM vacancies.	Weekly-quarterly	Postings, AI-mention share, vacancy trends, sector demand.

Tier	Sources	Latency	Role
Tier 4 — Displacement & event	Challenger, Gray & Christmas (30-industry job-cut table + AI-cited counts), Eurofound European Restructuring Monitor, Layoffs.fyi context, WEF Future of Jobs.	Monthly-annual	Sector layoffs, AI-attributed cuts, narrative, alternative attributions.

Wired live sources (v3, Aug 2026). Measured pairs pull weekly from: Indeed Hiring Lab (exposed-posting index and AI-mention share; US/UK/EU/AU), Adzuna (India and APAC-Singapore postings and AI-term share, plus the AI-skill salary premium for all six regions; requires the free API key configured as a repository secret), BLS/ONS/Eurostat/ABS/ILOSTAT (early-career unemployment delta, all six regions), Challenger Gray & Christmas (US AI-cited layoffs), ONS redundancies (UK all-cause proxy), BLS CPS youth employment (US hire-rate proxy), Eurostat recent-graduate employment (EU graduate proxy), and the Anthropic Economic Index (augmentation share, all six regions). Proxy-fed tiles are renamed on the dashboard to say exactly what they measure, and only while the proxy value is live. Where no open machine-readable source exists anywhere (net job creation; AI-attributed layoffs outside the US), the tile stays honestly modelled. The full per-KPI review lives in `SOURCE_REVIEW_2026-08.md` in the repository.

7.1 Refresh cycle

The site auto-refreshes every Monday at 06:00 UTC via a GitHub Actions workflow. It runs `scripts/update_data.py` (KPI refresh: new rows in `data/historical.csv`, a frozen snapshot in `data/snapshots/`, a rewritten `data/current.json`) followed by `scripts/update_sectors.py` (sector postings, employment, vacancies and layoffs into `data/sectors.json` plus the `data/sector_series.csv` archive). Quarterly workflows rebuild the occupation models and the sector exposure matrices. Netlify auto-deploys on every commit.

7.2 Open data

Every value the dashboard has ever shown is downloadable: `historical.csv` (all KPIs, all regions, all weeks), the weekly JSON snapshots, `current.json`, `sectors.json` (the sector layer: exposure, signals, provenance), `sector_series.csv` (append-only sector signal archive) and `uk_occupations.json` (the 412-occupation UK model). Snapshots are never edited after the fact — if a source revises a figure, the new value enters a future snapshot, not the past one. This makes the data store an audit trail as much as a source.

8. Limitations and caveats

8.1 AI washing

Layoffs.fyi indicates roughly 48% of 2026 tech layoffs are "AI attributed" by the employer, while a stricter count puts the genuine-automation figure closer to 20%. The Hill and Crunchbase both quote analysts arguing that AI announcements are partly a cover for reallocating budgets to AI infrastructure, not for net headcount reduction caused by automation. Every AI-attributed layoff event in this tracker carries an attribution-confidence score (1 to 5) so reports can be filtered by stringency.

8.2 Survey lag and sampling

Anthropic Economic Index releases are quarterly and based on a one-week sample window. BLS, ONS, NSO, and Eurostat LFS data publish with a one to three month lag. The dashboard surfaces the freshness timestamp on every tile so users know what is live and what is structural.

8.3 Country-brief comparability

Anthropic country briefs measure usage by Claude users in that country, not the country's workforce. Cross-country absolute comparisons of Observed Exposure are misleading. The within-country quartile rank is the only safe comparable.

8.4 Scraping legality

LinkedIn's terms prohibit unauthorised scraping; only the licensable LinkedIn Economic Graph feed is used. Indeed Hiring Lab, Seek Employment Trends, Naukri JobSpeak, and equivalent aggregators provide ToS-compliant aggregated feeds. The pipeline never stores raw job postings; it stores aggregated metrics only.

8.5 Causal modesty

Even with the best current measures, the differential unemployment signal in the US is statistically indistinguishable from zero. This tool is built to detect movement as it emerges, not to assert causation prematurely. Every monthly and yearly report includes a section on alternative explanations – trade, policy, business cycle, demographic shift – that could account for the same observation.

8.6 Derived layers, proxies and parsed sources

The AI Pressure Index, pillar signals and sector pressure are **presentation-layer derivations**: they add no new measurement, only fixed, documented calibration bands and weights over the underlying series, and their full lineage is visible in Deep view. Several tiles are honest proxies renamed on the dashboard (UK redundancies for layoffs, US youth employment for the hire rate, EU recent-graduate employment for graduate postings). Two sector signals are parsed from published text rather than an API – the Challenger industry table (PDF) and Naukri JobSpeak (report copy) – and are therefore brittle to format changes; a failed parse carries the previous value forward rather than inventing one. Sector layoff counts are **all-cause**, not AI-attributed: attribution exists only at the headline Challenger AI-cited figure. Where statistics agencies publish only industry sections, sectors sharing a section (banking and insurance; IT and telecom/media) share an exposure score, marked "section-level" on the dashboard.

9. References

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Prepared June 2026; revised 18 August 2026 — v2.0. This revision documents the Simple/Deep reading levels, the derived AI Pressure Index and pillar signals, the sector pulse across all six regions including the layoffs layer, and the v3 measured data sources. Updates to this manual follow major methodological revisions.